ESP32 WiFi LED Control Project

MicroPython • ESP32 • WiFi • Smart Home Basics

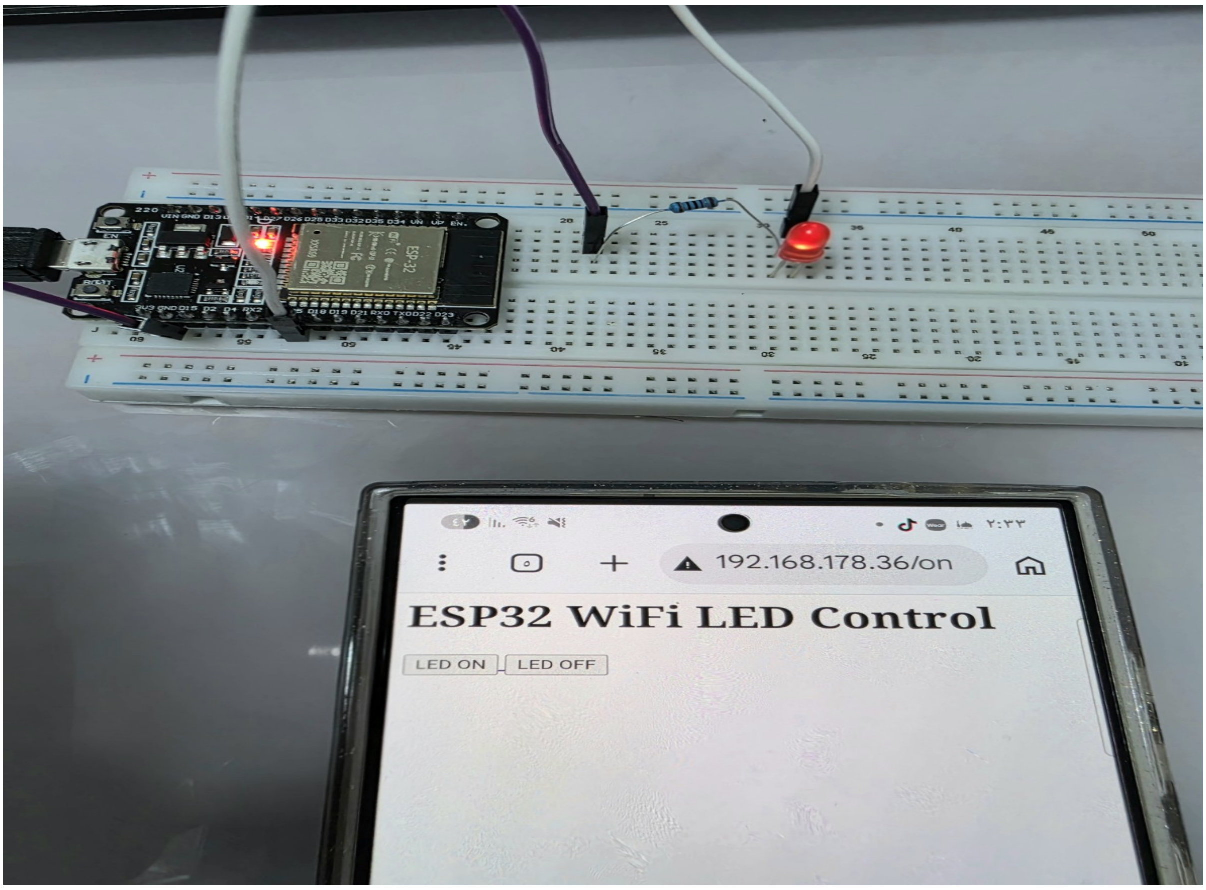


Figure 1: ESP32 WiFi LED Control using Smartphone Browser

Deutsche Dokumentation

Dieses Projekt zeigt die Steuerung einer LED über WiFi mit einem ESP32 und MicroPython. Der ESP32 verbindet sich mit einem WLAN-Router und erstellt einen kleinen Webserver. Über das Smartphone kann die LED direkt im Browser ein- und ausgeschaltet werden.

Verwendete Komponenten

Komponente	Beschreibung
ESP32 DevKit V1	Mikrocontroller-Board mit WiFi
LED	Visuelle Ausgabe
Widerstand	LED-Schutz
Breadboard	Schaltungsaufbau
Jumper Kabel	Elektrische Verbindungen
MicroPython	Programmiersprache

Verdrahtung

GPIO5 → LED → Widerstand → GND

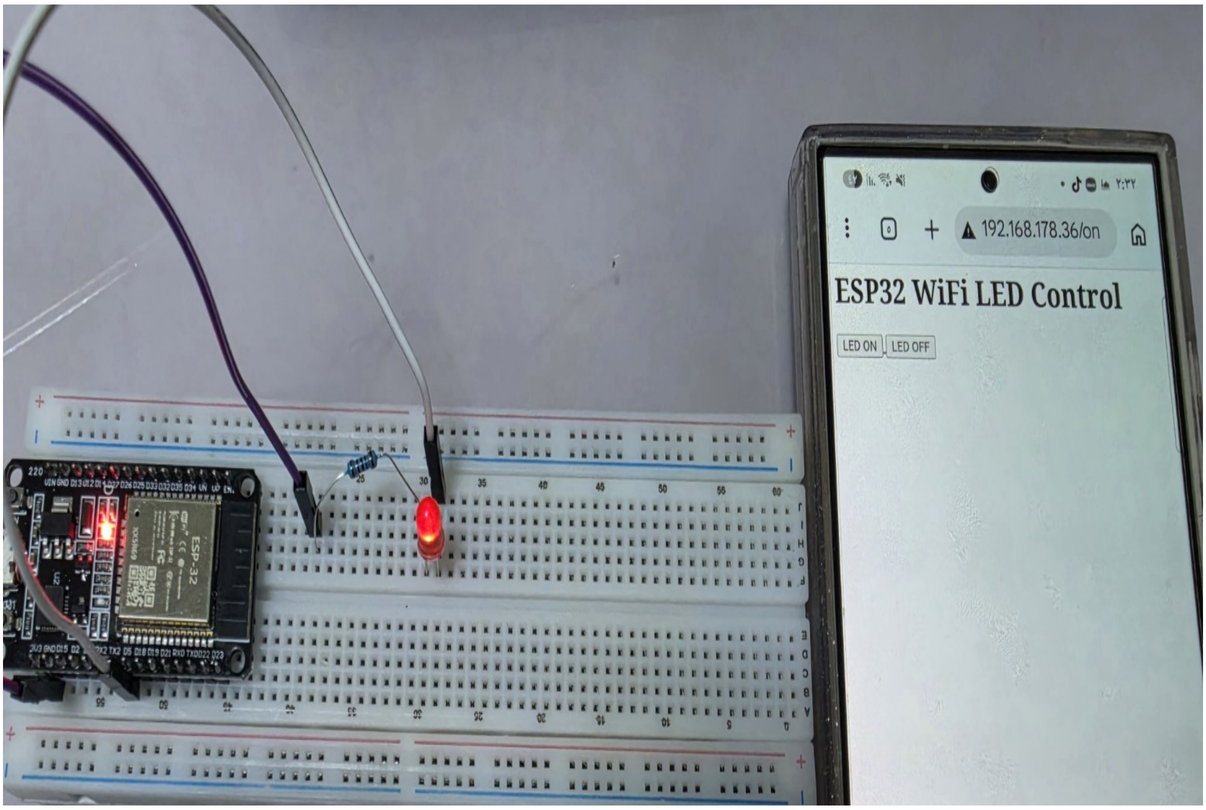


Figure 2: LED Control over WiFi

Programmcode

```
import network
import socket
from machine import Pin

ssid = 'Your_WiFi_Name'
password = 'Your_WiFi_Password'

led = Pin(5, Pin.OUT)

wifi = network.WLAN(network.STA_IF)
wifi.active(True)
wifi.connect(ssid, password)

while not wifi.isconnected():
    pass

addr = socket.getaddrinfo('0.0.0.0', 80)[0][-1]

server = socket.socket()
server.bind(addr)
server.listen(1)

while True:
    client, addr = server.accept()

    request = client.recv(1024)
    request = str(request)

    if '/on' in request:
        led.on()
    if '/off' in request:
```

```
    led.off()

html = '''
<html>
<body>
<h1>ESP32 WiFi LED Control</h1>
<a href="/on"><button>LED ON</button></a>
<a href="/off"><button>LED OFF</button></a>
</body>
</html>
'''

client.send(html)
client.close()
```

Projekt starten

1. ESP32 mit dem Computer verbinden.
2. Den Code als **main.py** speichern.
3. WLAN-Daten eingeben.
4. Das Programm starten.
5. Die IP-Adresse im Browser öffnen.
6. LED über die Webseite steuern.

English Documentation

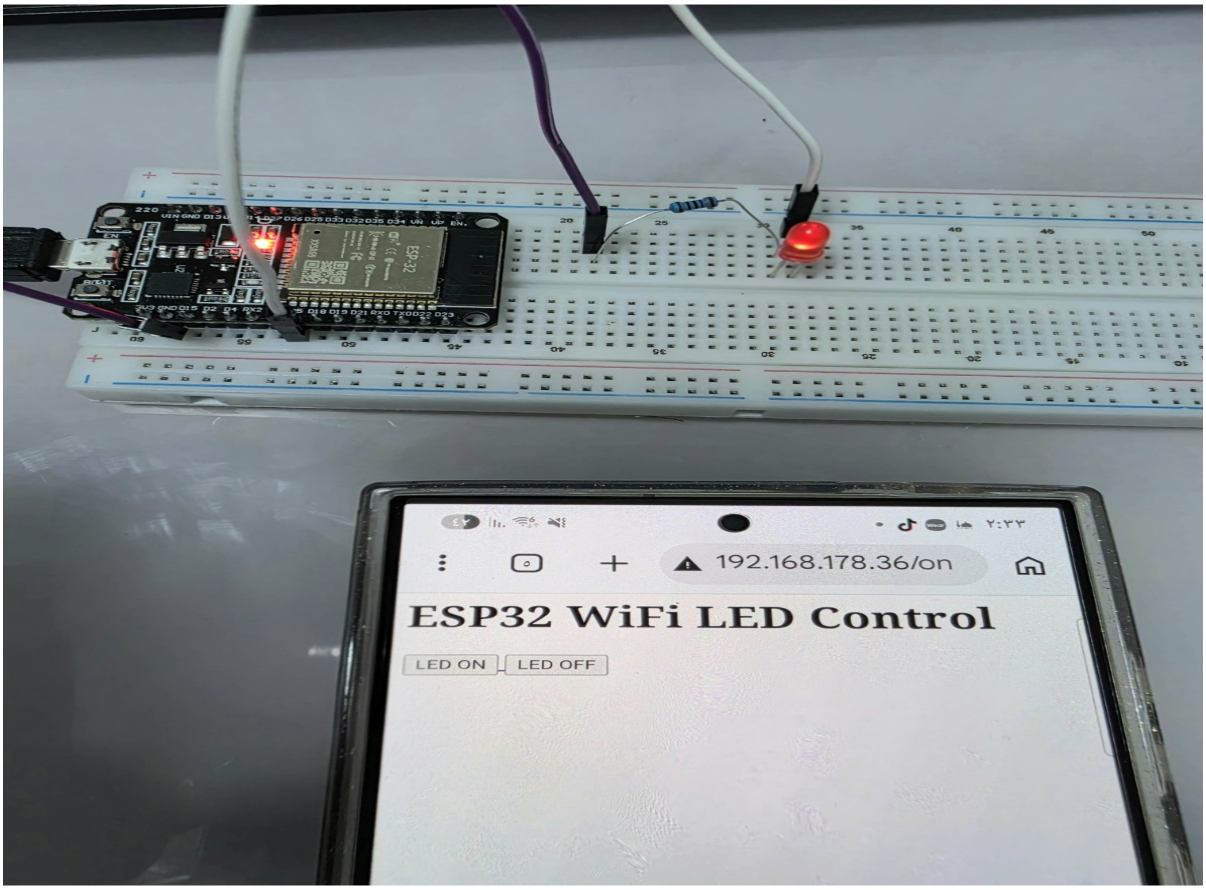
This project demonstrates controlling an LED over WiFi using ESP32 and MicroPython. The ESP32 connects to a WiFi router and creates a small web server. Using a smartphone browser, the LED can be turned ON and OFF remotely.

Components Used

Component	Description
ESP32 DevKit V1	Microcontroller board with WiFi
LED	Visual output
Resistor	LED protection
Breadboard	Circuit prototyping
Jumper Wires	Electrical connections
MicroPython	Programming language

Circuit Wiring

GPIO5 → LED → Resistor → GND



Program Code

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    </html>
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    client.send(html)
    client.close()
```

How to Run the Project

1. Connect the ESP32 to the computer.
2. Save the code as **main.py**.
3. Enter the WiFi credentials.
4. Run the program.
5. Open the IP address in the browser.
6. Control the LED from the webpage.

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